OPTIMIZATION OF HURRICANE EVACUATION NETWORKS

Network Optimization • Scenario Analysis • Sensitivity Analysis • Decision Support

EXECUTIVE SUMMARY

Objective

Develop a network optimization model to evaluate hurricane evacuation strategies across major Florida population centers and shelters

Methods

- Linear Programming
- Network Flow Optimization
- Capacity-Constrained Routing
- Congestion Minimization
- Staggered Evacuation Scheduling
- Scenario and Disruption Analysis

Outcome

The model provides a decision-support framework for evaluating evacuation plans, infrastructure disruptions, and shelter allocation strategies

OVERVIEW

Models Evaluated

- Distance Minimization
- Travel Time Minimization
- Capacity-Constrained Evacuation
- Distance + Capacity Optimization
- Congestion Minimization
- Manual Staggered Evacuation
- Optimized Staggered Evacuation

Additional Analyses

- Shelter Utilization
- Road Utilization
- Sensitivity Analysis
- Infrastructure Disruption Scenarios
- Bottleneck Identification

This project investigates how operations research techniques can improve hurricane evacuation planning in Florida.

A transportation network was constructed using population centers, shelter locations, travel times, and road capacities. Multiple optimization models were developed to evaluate evacuation performance under different objectives and operational constraints.

DATA

Road Network Data

- Distances and travel times were collected using Apple Maps by identifying direct routes between connected cities
- Each route was represented as an edge within the transportation network
- Road Capacities were formulated by number of people per car * evacuation time * lanes * road car support
- Number of people per car was assumed to be 2.5, evacuation time was changeable but assumed to be 12 hours, lanes was found to by Apple Maps and is the smallest number found, and road car support was found by what if it was mainly highway, freeway, or a local road

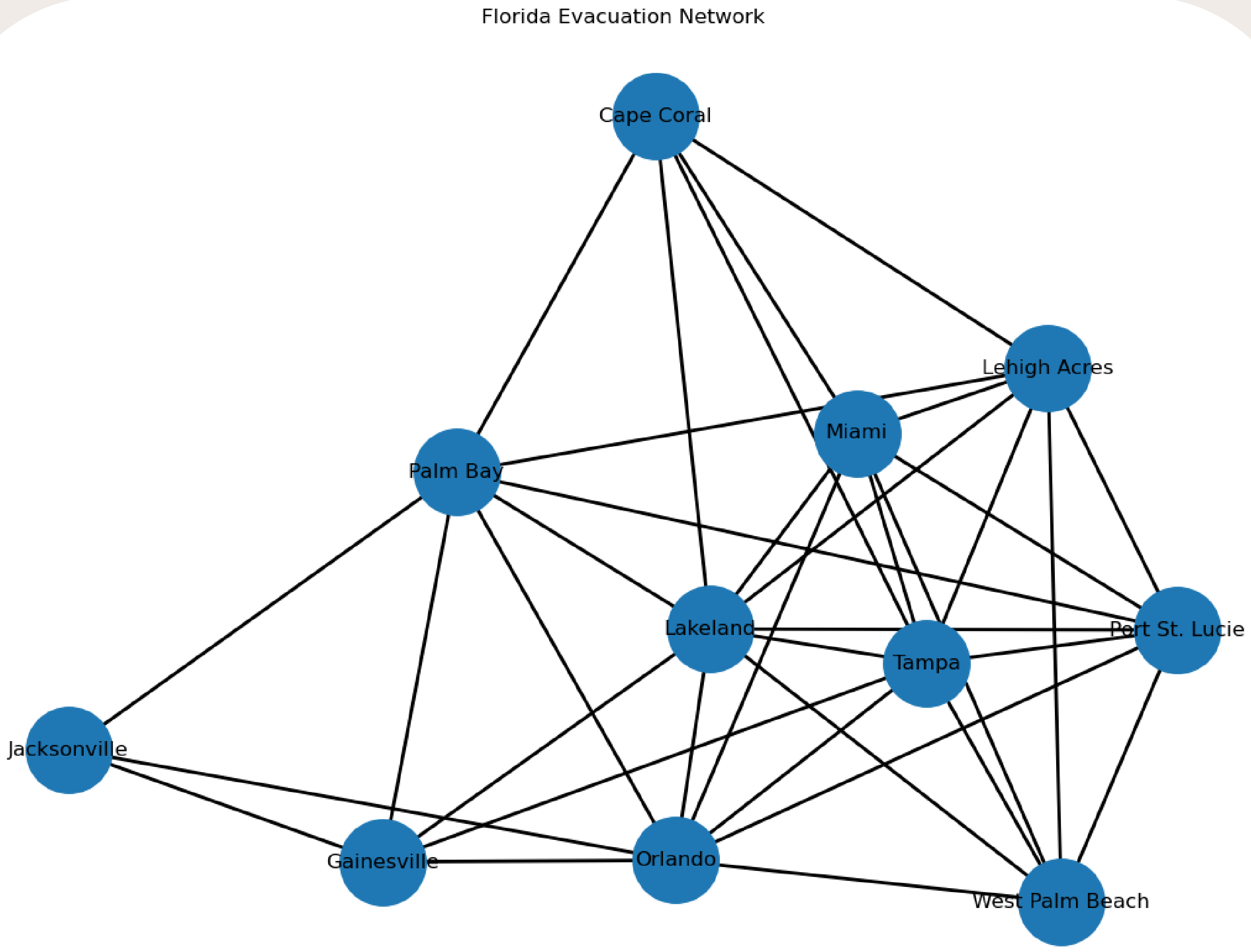
Population Data

- City populations were obtained from the 2020 U.S. Census

Shelter Capacities

- Shelter capacities were estimated using rounded city population values to represent available evacuation capacity

Origin Cities	Miami
	West Palm Beach
	Port St. Lucie
	Lehigh Acres
	Cape Coral
	Palm Bay
Shelter Cites	Orlando
	Lakeland
	Tampa,
	Jacksonville
	Gainesville



PROBLEM STATEMENT

- How should evacuees be routed?
- How does congestion affect evacuation?
- What happens when infrastructure fails?
- Which shelters and roads are critical?

MATHEMATICAL FORMULATION

Decision Variable

x_{ij} – number of people traveling from node i to node j

Objective

$\min \sum c_{ij} * x_{ij}$

Where c_{ij} = travel distance or travel time on road (i, j)

The objective is to minimize total evacuation cost (distance or time)

Constraints

- Flow Conservation
  - $\sum_j x_{ij} - \sum_j x_{ji} = P_i$
  - The net outflow from each origin city must equal it's population
  - All evacuees must leave their origin city
- Shelter Capacity
  - $\sum_i x_{ij} - \sum_i x_{ji} \leq C_j$
  - The number of evacuees assigned to a shelter can't exceed its capacity
- Road Capacity
  - $x_{ij} <= R_{ij}$
  - The flow on road (i, j) can't exceed the road's capacity

MODELS

Distance

- Minimizes total travel distance (miles)
- Doesn't include road capacity constraints
- Routes evacuees using the shortest-distance paths

Time

- Minimizes total travel time (hours)
- Doesn't include road capacity constraints
- Routes evacuees using the fastest paths

Capacity

- Minimizes total travel time
- Includes road capacity constraints
- Forces alternate routes when roads reach capacity
- Represents congestion effects on evacuation

Distance + Capacity

- Minimizes total travel distance
- Includes road capacity constraints
- Balances shortest routes with roadway limitations

Min Congestion

- Minimizes the maximum road utilization
- Doesn't optimize travel distance or time
- Distributes evacuees across the network to reduce bottlenecks and heavily used roads

Manual Stagger

- Minimizes total travel time with road capacities
- Evacuation occurs in three predefined waves
- Each wave is assigned a fixed percentage of evacuees and shelter capacity
- Designed to reduce congestion by spreading departures over time

Stagger

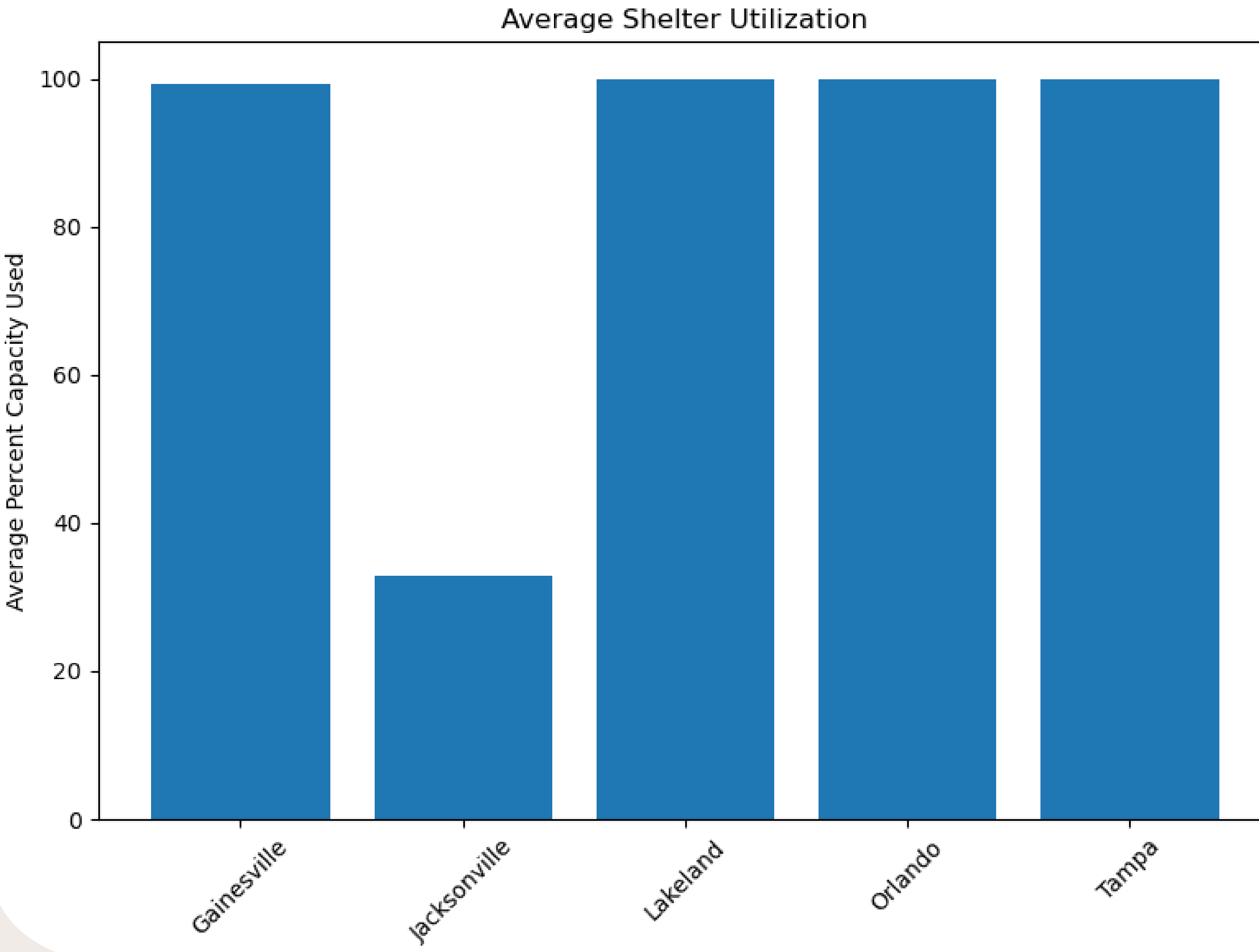
- Minimizes total travel time with road capacities
- Evacuation occurs in multiple waves with flow optimized simultaneously across the waves
- Optimization determines how the evacuees are distributed throughout the staggered evacuation schedule

Model	Total Person-Hours	Total Distance-Miles	Max Congestion	Active Roads
Distance	4.59e+6	2.67e+8	303%	10
Time	4.48e+6	2.69e+8	463%	10
Capacity	4.79e+6	2.85e+8	100%	23
Distance + Capacity	4.83e+6	2.84e+8	100%	25
Min Congestion	5.49e+6	3.24e+8	79%	31
Manual Stagger	4.518e+5	2.711e+8	100%	15
Stagger	4.515e+5	2.709e+8	100%	15

MODEL COMPARISON

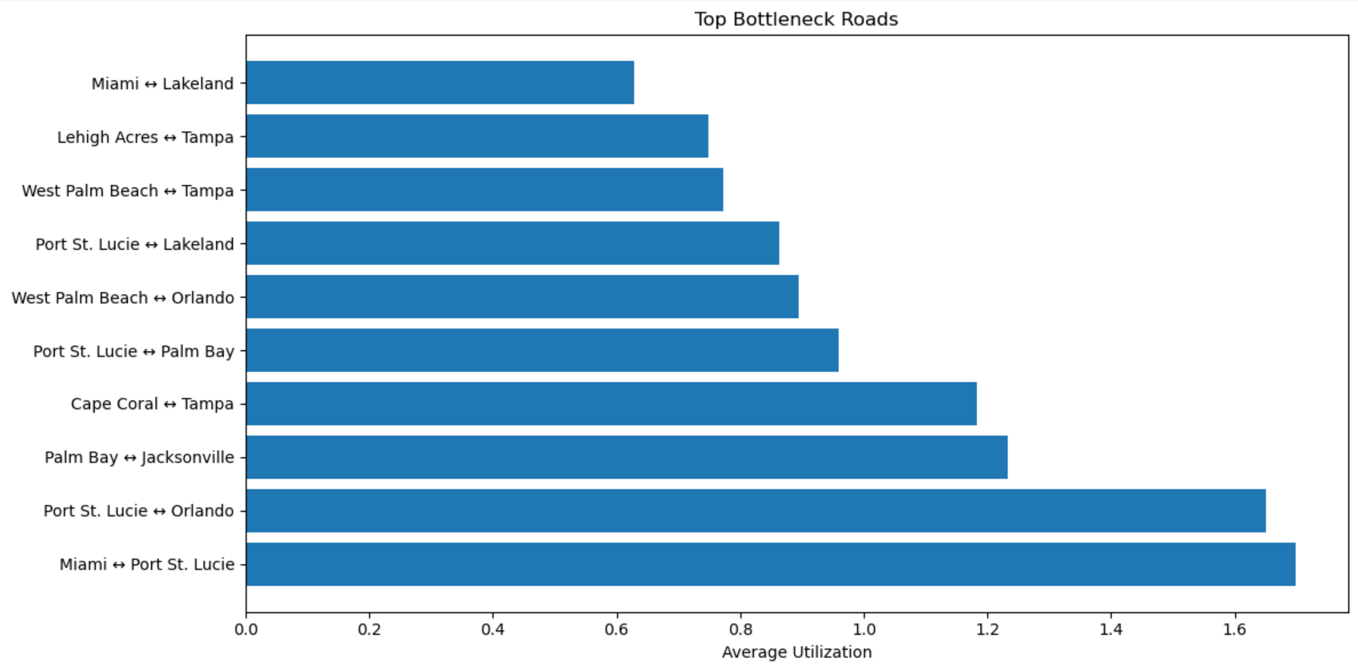
SHELTER UTILIZATION

- Tampa, Orlando, and Lakeland consistently reach capacity across all optimization models
- Jacksonville and Gainesville primarily act as overflow shelters once the closer shelters become full.
- Shelter assignments remain largely consistent across optimization approaches
- The Manual Stagger model is the only model that noticeably changes shelter allocation, shifting approximately 66,500 evacuees from Gainesville to Jacksonville
- Results suggest that shelter capacity, rather than route selection, is the primary driver of final shelter occupancy

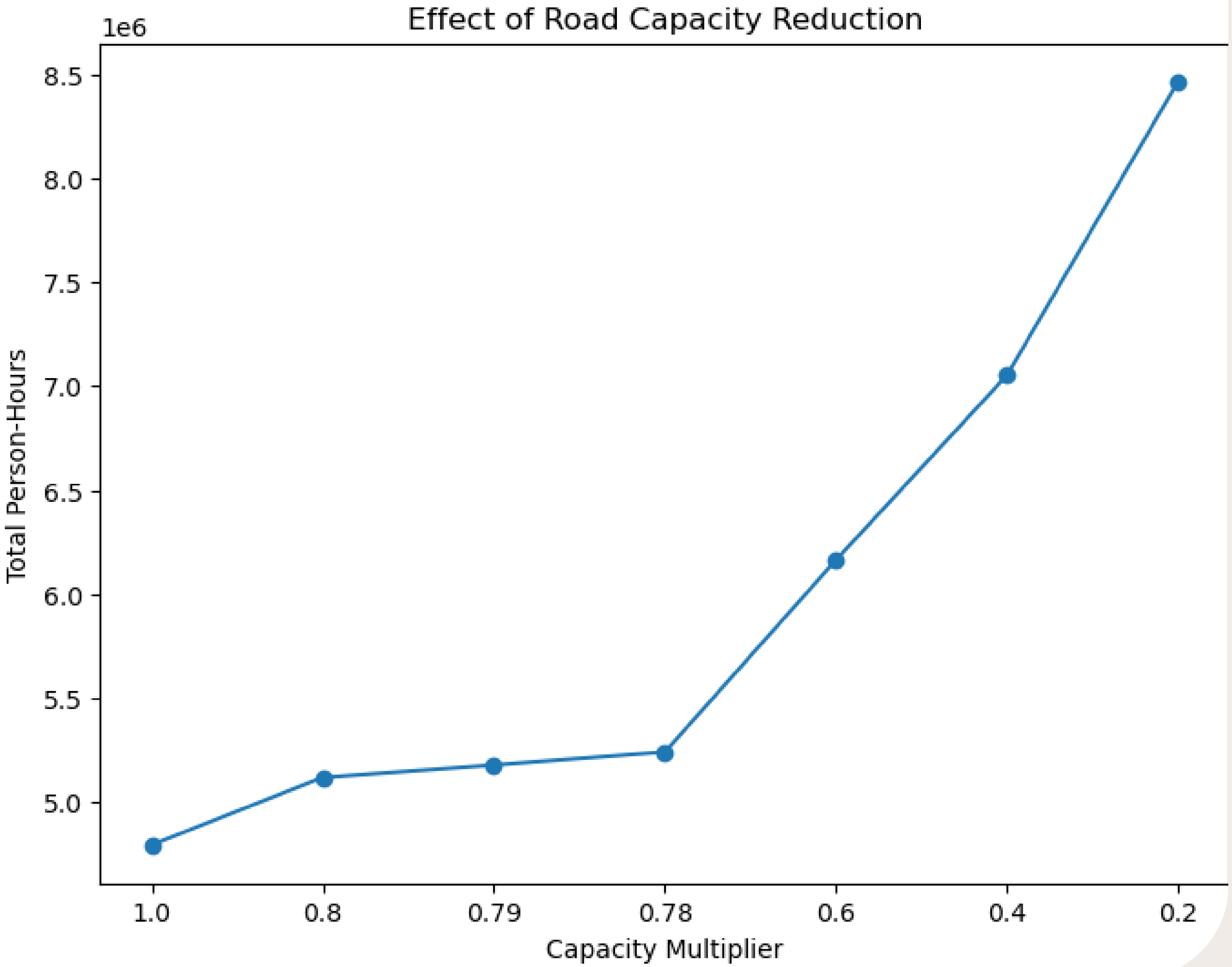


ROAD UTILIZATION

- The Distance and Time Models use 8 of the same roads with high utilizations and then 2 others with low capacities
- Road-capacity constraints distribute 8 high-capacity roads to be 100% capacity across about 15 roads
- Min Congestion model balances the network utilization with 79% on the 100% capacity and gets use out of the non-used roads
- Staggered evacuation reduces congestion across many roads and reflects the original models with number of roads but with capacity constraints
- Minimizing travel time increases congestion, while congestion-focused strategies require longer travel distances and times



- Routes connecting Orlando, Tampa, and Lakeland consistently experience the highest utilization levels
- These corridors serve as major transfer routes between evacuation origins and shelter locations
- When shelter capacities become constrained, utilization shifts toward routes leading to Jacksonville and Gainesville
- The route connecting Miami, Port St. Lucie, and Palm Bay have the highest averages as that's where most of the evacuees are leaving, and the areas are close together using similar roads



SENSITIVITY ANALYSIS

- Road Capacity was scaled using multipliers of 1.0, 0.8, 0.79, 0.78, 0.6, 0.4, and 0.2
- As road capacity decreased, the model required evacuees to take longer and less efficient routes, increasing total travel time and distance
- The network remained feasible at 0.79, but at 0.78 became infeasible, indicating a critical threshold for successful evacuation
- Maximum road congestion increased dramatically, rising from 1.0 at 0.79 to over 43 at 0.2
- Shelter usage shifted as at 0.4 Gainesville was at 276% capacity while Jacksonville was at 3%
- The trend continued at 0.2 with Tampa and Orlando sending people to Gainesville instead (670%)
- The results suggest the evacuation network is highly sensitive to reduction in roadway capacity and has limited redundancy once key routes become constrained

DISRUPTION SCENARIOS

- None is the base capacity model
- Tampa Blocked reduces the capacity of the roads around Tampa and tests the loss of a major shelter
- Tampa Evacuates has half the population of Tampa evacuate and the shelter capacity is halved
  - Was originally all of Tampa but this was infeasible
  - Tests a shelter loss with population increase
- Limited Orlando reduces the shelter capacity by one half
- Extra Orlando doubles the shelter capacity
- Blocked Orlando makes it where once you enter Orlando you can't leave (outflow = 0)
  - The Orlando scenarios tested all possible events for a transportation node
- Limited Gainesville halves the shelter capacity of Gainesville and tests the lost of an overflow shelter
- Miami Blocked halves the capacity of the roads leaving Miami and tests the loss of evacuations routes
- Extra Population scales the population in Palm Bay, Palm Beach, and Lehigh Acres by 1.25
  - Tests what would happen with a higher population
- One Way Roads makes all roads one way and doubles the road capacity
  - Tests what would happen with more road capacity

Disruption	Total Hours	Total Miles	Max. Congestion	Active Roads
None	4.79e+6	2.85e+8	100%	23
Tampa Blocked	5.25e+6	3.10e+8	100%	28
Tampa Evacuates	6.96e+6	4.17e+8	784%	20
Limited Orlando	5.25e+6	3.14e+7	194%	21
Extra Orlando	4.33e+6	2.54e+8	100%	24
Blocked Orlando	4.93e+6	2.94e+8	207%	22
Limited Gainesville	4.87e+6	2.90e+8	100%	25
Miami Blocked	4.90e+6	2.89e+8	100%	29
Extra Population	5.30e+6	3.14e+8	104%	25
One Way Roads	4.50e+6	2.70e+8	100%	13

BEST CASE SCENARIO

Best Total Hours

- One Way Roads
- 4.50 Million hours which is closer to the model with no capacity

Best Congestion

- One Way Roads reaches 1 on 5 out of 13 roads, the least amount of times
- Also uses the least number of roads, focusing the congestion on those 5 roads

Least Jacksonville Usage

- Extra Orlando with 162000 people compared to the usual 311000
- The extra shelter space in Orlando makes the over-flow less needed

Findings

- Making each road go one way creates the best-case scenario by decreasing transportation travel time and distance alongside congestion
- Orlando opening more shelter space is the second-best scenario by decreasing travel time and the overflow shelters

WORST CASE SCENARIO

Worst Total Hours

- Tampa Evacuates
- 6.96 Million hours which is a 45% increase from the baseline

Worst Congestion

- Tampa Evacuates with Max Congestion = 7.84 on Tampa to Jacksonville
- Miami to Orlando increase to 4.51 and Orlando to Gainesville to 4.19 (a previously unused route)
- Tampa to Lakeland rises slightly to 1.01
- Causes bottlenecks throughout the network that are even worse than when there’s no road capacity constraints

Most Jacksonville Usage

- Tampa Evacuates with 888500 people compared to the usual 311000

Findings

- Tampa functions as both a major shelter and transportation hub
- Removing Tampa and adding it as an evacuation hub causes the greatest degradation in the system performance
- The model becomes infeasible as there is no way to keep congestion under 1, creates the worst travel time (total and average), and the worst travel distance by 100 million miles compared to the baseline

KEY FINDINGS

Road Capacity is the primary driver of evacuation performance

- Increasing road capacity significantly reduced evacuation time and congestion
- The network became infeasible when road capacity fell below approximately 79% of baseline levels
- Evel slight changes around key cities can change travel time by thousands of hours

Tampa & Orlando are critical nodes

- Orlando is important as a transportation hub to get to other cities and blocking that led to increases in congestion making the model infeasible
- Disruptions involving Tampa produced the largest increases in travel time, distance, and congestion
- The reverse of Tampa, the largest decreases in travel time and congestion relating to change in a shelter stemmed from Orlando

Jacksonville functions as the primary overflow shelter

- When major shelters reached capacity, evacuees rerouted to Jacksonville
- Jacksonville usage increased substantially under disruption scenarios

Congestion reduction requires longer travel routes

- The min congestion model produced the lowest bottleneck utilization but came at the cost of higher travel distances and total hours

Staggered evacuations improve network performance

- Staggering demand reduced simultaneous pressure on transportation infrastructure
- More balanced route utilization was achieved compared to non-staggered evacuations

The network is resilient to moderate disruptions

- Major changes such as population, major shelter capacity, and major road capacities lead to infeasible solutions
- Severe disruptions increased travel time and congestion rather than complete network failure

ASSUMPTIONS

- All residents evacuate
- Travel times and road capacities are known and fixed
- Shelter capacities remain constant
- Evacuees follow optimized routes
- No weather, accidents, or human behavior effects are modeled
- Population and shelter capacities are represented as aggregate flows
- Only selected Florida Cities exist as origins and shelters
- Only modeled routes are available, and background traffic is not considered

RECOMMENDATIONS

- Use a capacity-constrained, staggered evacuation plan
- Open all lanes on major highways going towards shelters
- Prioritize maintaining access to Tampa and Orlando
- Preserve roadway capacity above 80% of normal operating conditions
- Use Jacksonville as the primary overflow shelter during large-scale evacuations

CONCLUSIONS

The evacuation network can successfully relocate all evacuees under normal conditions

- All optimization models produced feasible evacuation plans while satisfying shelter and roadway constraints
- However, it is quite easy to modify the conditions to make the network infeasible

Roadway capacity is more critical than shelter capacity

- Transportation infrastructure was the primary limitation of evacuation performance

No single optimization objective is best in every situation

- Distance models minimize travel distance
- Time models minimize travel time
- Congestion models reduce bottlenecks
- Staggered models distribute demand over time
- Each approach involves tradeoffs between efficiency and network resilience

Staggered evacuations are an effective congestion-management strategy

- Distributing departures over time reduced pressure on critical routes and allowed more cars on it
- Staggering offers a low-cost method to improve evacuation performance

FUTURE WORK

1. Dynamic traffic assignment
2. Real-time congestion updates
3. Additional shelter locations
4. Weather-dependent road closures
5. Multi-state evacuation scenarios
6. Integration with GIS mapping tools